

LECTURE XX

Lecture Title On Two Lines

One or two sentences saying what the lecture is actually about — the question it answers, not the topics it covers.

Applied PINN for Energy · Aalborg University

keyword · keyword · keyword · keyword

[opening figure]

Where This Sits

WHAT CHANGES, AND WHAT DOES NOT

First paragraph. Full sentences — this is a deck you read from, and the recording script is generated from the notes, not from bullets.

Second paragraph. Aim for three or four here; more than five and the column runs past the fold.

Third paragraph. The last one should land the point rather than trail off.

A KEY/VALUE CARD

term	what it means here
term	what it means here
term	what it means here
term	four rows fits; five is tight

A PROSE CARD

Cards take smaller type — pass 9.5 as the last argument to paras().

Two short paragraphs is the most that fits comfortably.

The Governing Equation

WHAT THE EQUATION SAYS

Say in words what the equation asserts, before showing it. A student who cannot state it in a sentence has not understood it.

Then what each term is, and which ones are known.

Then why it is usable as a residual rather than only as something to solve.

[addEq(s, "governing", M, 4.30, 6.3) — see LXX_equations.py]

a caption naming the equation and where it comes from

A DIAGRAM BELONGS HERE

[figure]

THE SUBTLETY WORTH STOPPING ON

The one assumption, sign convention or boundary condition that quietly ruins the result if it is missed. Every lecture has one. Find it and put it here.

Two Ways to Do It

YOU HAVE SEEN THIS BEFORE

A full-width callout across the top is how the course flags a callback to an earlier lecture. Name the lecture and what transferred.

OPTION A — THE EASY ONE

What it is, in two sentences.

What it costs — and be honest, because the reason students choose wrongly is that the cost was never stated.

When to use it.

OPTION B — THE ONE THAT WORKS

What it is.

Why it is stronger.

When it is worth the extra effort.

One italic line at the bottom giving the verdict. Students photograph this line.

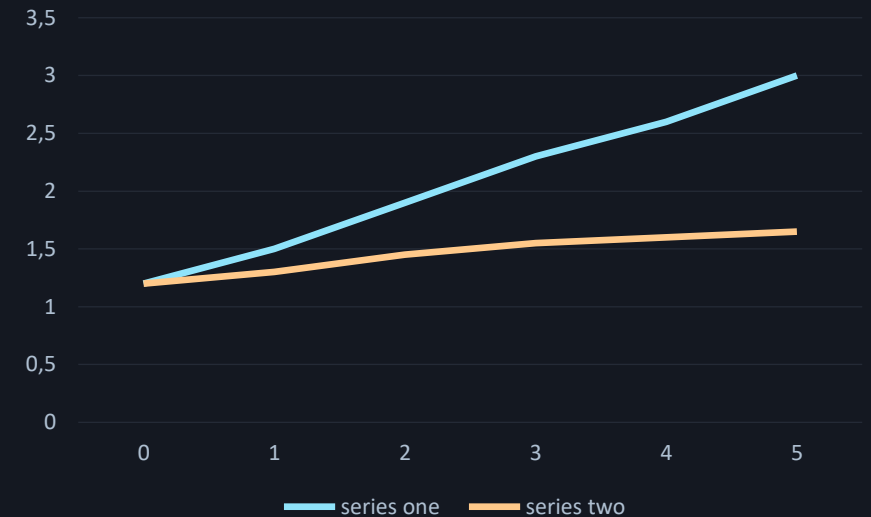
What the Numbers Show

READ THE AXES BEFORE THE CURVE

Say what is plotted and against what, then what the shape means.

If the data is illustrative rather than measured, say so on the slide — the caption below the chart is the place.

A TITLE FOR THE CHART



Illustrative. Say so when it is.

Failure Modes

IT TRAINS BUT IS WRONG

The quiet failure. Loss goes down, the picture looks right, and the answer is not a solution. Say how to detect it.

IT WILL NOT TRAIN

The loud failure, and the easier one. Usually scaling or a weight.

IT WORKS AND DOES NOT GENERALISE

The one that survives to the report. Held-out conditions, not held-out samples.

WHAT TO CHECK, IN ORDER

first	the cheapest check that could rule it out
then	the next cheapest
then	the one that takes a rerun
last	the one that takes a rewrite

Ordering the checks by cost is the actual engineering content of this slide.

Ex_LXX — What You Will Build

THE SHAPE OF THE WORK

One paragraph on what the exercise asks and why it is the right question for this lecture.

One on what is provided versus what they write.

One on what the deliverable is.

THE NOTEBOOKS

00	reference — read only, generates ground truth
01	the baseline, with TODOs
02	the method, with TODOs
03	comparison and report

THE QUESTION THAT CARRIES THE MARKS

State it on the slide. The marks are for the reasoning, not the number.

Key Takeaways

1	It is the same machinery	Name the continuity with an earlier lecture.
2	The thing that bites	The assumption or scaling that ruins it.
3	The trade	What you gave up to get what you got.
4	The expensive part	Where the compute or the effort actually went.
5	Where the method belongs	Not faster — different. Say what only this can do.

Next — LXX+1: the following lecture, in one clause.

References and Further Reading

THE CLASSICAL SIDE

Author (year), Title — one clause saying what it is for.

Author (year), Title — and why this one rather than another.

Author (year), Title.

THE PHYSICS-INFORMED SIDE

Raissi, Perdikaris & Karniadakis (2019), J. Comput. Phys. 378, 686–707 — the formulation used throughout this course.

Liu (2025), PINN with Python — the implementation conventions the exercises follow.

READ IN THIS ORDER

if you are new to this the gentlest source first

if you know the field go straight to the paper

for the exercise the run guide

for next week the chapter that sets it up

Say plainly whether anything here is behind a paywall the library does not cover.

NEXT

LXX+1 — The Next Lecture Title

Two sentences on what the next lecture does with what this one built.

BEFORE THE NEXT SESSION

run which notebooks

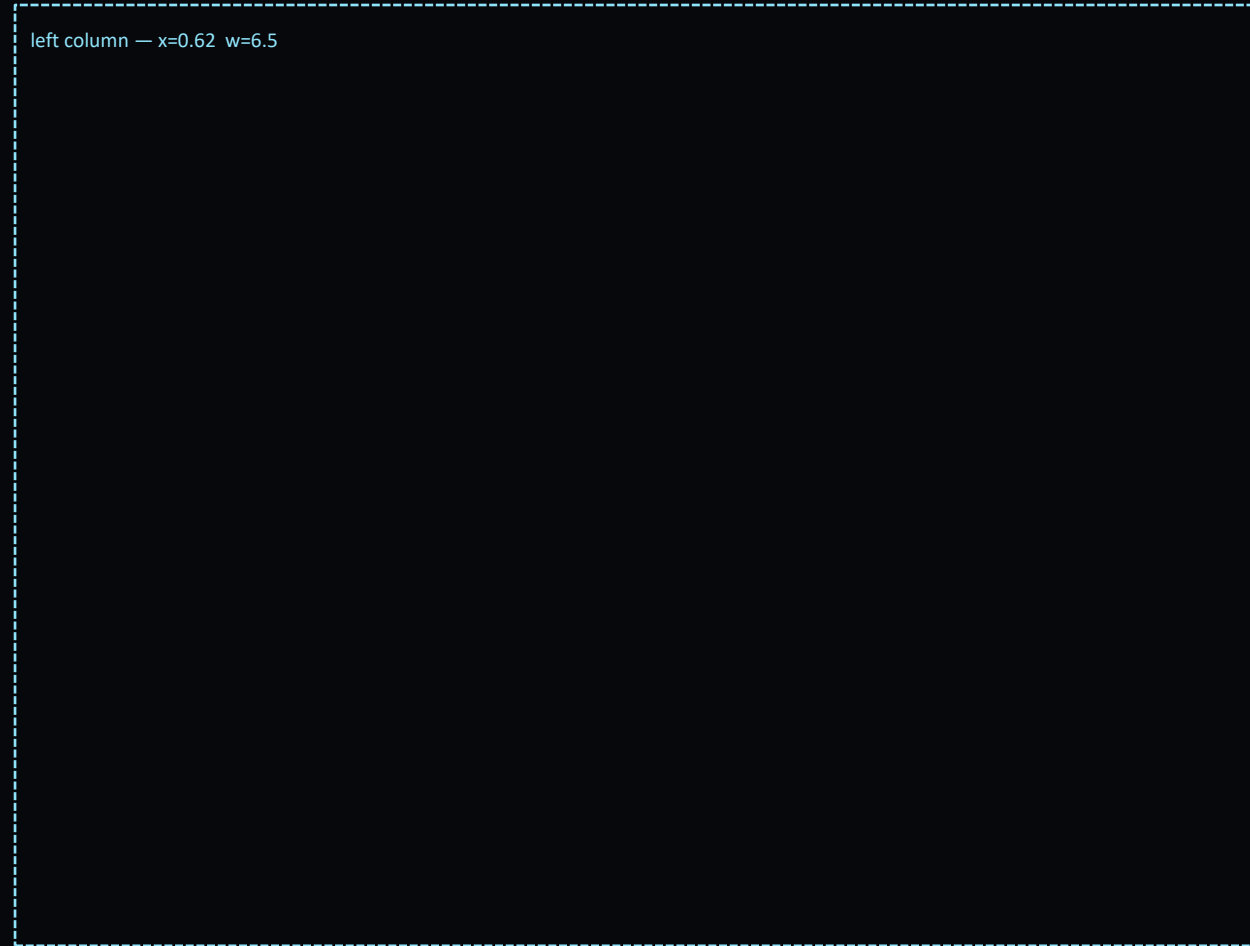
bring which numbers

think about the question that opens next time

read the chapter, if they can

Say why the preparation matters, concretely. 'Read chapter 4' is ignored; 'arrive without a baseline and you will spend the session making one' is not.

Layout Reference — delete this slide



Canvas 13.33×7.5 in. M = 0.62 (left margin), CW = 12.09 (content width).

Kicker sits at $y=0.34$, title at $y=0.66$, the rule and footer at $y=6.92 / 7.00$ — all drawn by base().

Applied PINN for Energy | LXX — Lecture Title Goes Here

Content starts at $y=1.50$ and must end by $y=6.80$.

Helpers and Palette — delete this slide

EVERY HELPER THE THEME GIVES YOU

<code>base(s, {kicker,title,n,src})</code>	kicker, title, rule, footer, page number
<code>card(s, x,y,w,h [,accent])</code>	rounded panel; accent tints the border
<code>callout(s, x,y,w,h, LABEL, body, col)</code>	a card with a coloured label
<code>paras(s, x,y,w, [...] [,size])</code>	spaced body paragraphs, default 10.5pt
<code>sectionLabel(s, x,y,w, TEXT [,col])</code>	small tracked caps heading
<code>rows(s, x,y,w, [[k,v],...] [,col])</code>	two-column key/value list
<code>addEq(s, name, x,y,w [,containerW])</code>	an equation PNG from <code>src/eq/</code>
<code>CHART_BASE</code>	spread into <code>addChart</code> for course styling

THE PALETTE, AND WHAT EACH IS FOR

	CYAN	structure, callbacks, the default accent
	AMBER	attention, section labels inside cards
	ORANGE	a trap or a failure
	GREEN	a result, or the option that works
	PURPLE	a subtlety worth stopping on
	WHITE	titles and the one number that matters
	TXT	body prose
	MUTED	captions, secondary text